EECS 182 – Homework 1 Warm-Up Worksheet

Instructions. This worksheet is just for you; not for submission. It is meant to refresh exactly the math and ideas Homework 1 leans on from HW0 and Discussion 1. Try to be able to do this whole sheet without notes; if you get stuck, that flags a topic to review.

1 Vector and Matrix Calculus Refresher

Let $\boldsymbol{x}, \boldsymbol{c} \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$. We follow the convention that gradients of scalars with respect to column vectors are row vectors.

1.1 $f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{c}$

(a) What is the shape of $\dfrac{\partial f}{\partial \boldsymbol{x}}$ (scalar, vector, or matrix)?

(b) Compute $\dfrac{\partial f}{\partial \boldsymbol{x}}$ explicitly as a row vector in terms of $\boldsymbol{c}$.

1.2 $f(\boldsymbol{x}) = \|\boldsymbol{x}\|_2^2 = \boldsymbol{x}^\top \boldsymbol{x}$

(a) Rewrite $f(\boldsymbol{x})$ as $\sum_{i=1}^n x_i^2$.

(b) Compute $\dfrac{\partial f}{\partial \boldsymbol{x}}$ as a row vector and simplify it in compact matrix notation.

1.3 $\boldsymbol{f}(\boldsymbol{x}) = A\boldsymbol{x}$ (vector-valued)

(a) What is the shape of $\dfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}$ (the Jacobian)?

(b) Write the Jacobian explicitly in terms of A.

1.4 $f(\boldsymbol{x}) = \boldsymbol{x}^\top A \boldsymbol{x}$

(a) Expand $f(\boldsymbol{x})$ as a double sum $\sum_i \sum_j a_{ij} x_i x_j$.

(b) Compute the partial derivative $\dfrac{\partial f}{\partial x_i}$ for a fixed i.

(c) Stack your answers into $\dfrac{\partial f}{\partial \boldsymbol{x}}$ and write it cleanly in matrix form.

(d) Under what condition on A does this derivative simplify to $2\boldsymbol{x}^\top A$?

2 SVD, Least Squares, and Minimum-Norm Solutions

Let $X \in \mathbb{R}^{m\times n}$, $\boldsymbol{y} \in \mathbb{R}^m$, and take the full SVD

$$X = U\Sigma V^\top, \quad U \in \mathbb{R}^{m\times m},\ V \in \mathbb{R}^{n\times n},\ \Sigma \in \mathbb{R}^{m\times n}.$$

2.1 Overdetermined case ($m > n$, full column rank)

We solve

$$\min_{\boldsymbol{w}} \ \|X\boldsymbol{w} - \boldsymbol{y}\|_2^2.$$

(a) Using the normal equations, derive $\boldsymbol{w}^\star$ in terms of X and $\boldsymbol{y}$.

(b) Substitute $X = U\Sigma V^\top$ and simplify $\boldsymbol{w}^\star$ to the form

$$\boldsymbol{w}^\star = V\Sigma^+U^\top\boldsymbol{y},$$

where Σ^+ is the pseudoinverse diagonal matrix with entries $1/\sigma_i$. Show the algebraic steps.

2.2 Underdetermined case ($m < n$, full row rank)

We want the minimum-norm solution satisfying $X\boldsymbol{w} = \boldsymbol{y}$.

(a) Write the optimization problem that defines the minimum-norm solution.

(b) Derive the solution $\boldsymbol{w}_{\min}$ in terms of X and $\boldsymbol{y}$ using Lagrange multipliers or otherwise.

(c) Express $\boldsymbol{w}_{\min}$ in SVD form using U, Σ, V and show that it equals $V\Sigma^+U^\top\boldsymbol{y}$ as well.

2.3 Concept check

(a) In one sentence each, explain the meaning of “least-squares solution” versus “minimum-norm solution”.

(b) In which case ($m > n$ or $m < n$) are these solutions unique?

3 Ridge Regression Basics

Let $X \in \mathbb{R}^{n\times d}$, $\boldsymbol{y} \in \mathbb{R}^n$, and $\lambda > 0$. Consider ridge regression

$$L(\boldsymbol{w}) = \|\boldsymbol{y} - X\boldsymbol{w}\|_2^2 + \lambda\|\boldsymbol{w}\|_2^2.$$

3.1 Closed-form solution

(a) Take the gradient of $L(\boldsymbol{w})$ with respect to $\boldsymbol{w}$ and set it to zero.

(b) Solve for $\boldsymbol{w}_{\text{ridge}}$ in terms of X, $\boldsymbol{y}$, and λ.

3.2 SVD perspective and shrinkage

Let $X = U\Sigma V^\top$ as before.

(a) Substitute $X = U\Sigma V^\top$ into your formula for $\boldsymbol{w}_{\text{ridge}}$ and simplify until the only non-identity diagonal matrix is of the form diag(some scalar function of σ_i).

(b) What scalar factor multiplies each component of $U^\top \boldsymbol{y}$ in the V basis? Write it as a function of σ_i and λ.

(c) Describe what happens to directions with very small σ_i and with very large σ_i as λ grows.

3.3 Interpretation

(a) Explain in your own words why ridge regression is called "shrinkage".

(b) As $\lambda \to 0$, what does $\boldsymbol{w}_{\text{ridge}}$ tend to (assuming $X^\top X$ is invertible)?

(c) As $\lambda \to \infty$, what vector does $\boldsymbol{w}_{\text{ridge}}$ approach?

4 Scalar Gradient Descent Stability

Consider the scalar objective

$$L(w) = (y - \sigma w)^2, \quad \sigma > 0.$$

We run gradient descent with learning rate η:

$$w_{t+1} = w_t - \eta \left.\frac{dL}{dw}\right|_{w_t}.$$

4.1 Derive the update

(a) Compute $\frac{dL}{dw}$.

(b) Write the explicit recurrence w_{t+1} in terms of w_t, y, σ, and η.

(c) Identify the optimal solution $w^\star$ that minimizes L.

4.2 Error dynamics

Let $e_t = w_t - w^\star$.

(a) Derive a recurrence relation for e_{t+1} in terms of e_t.

(b) Find the common ratio $r(\eta, \sigma)$ such that $e_t = r^t e_0$. What is r?

4.3 Stability region

(a) For gradient descent to converge, what condition must $|r|$ satisfy?

(b) Use that condition to find the range of $\eta > 0$ for which the recurrence is stable. Express it as an inequality involving σ.

4.4 Convergence speed

(a) Suppose $|e_t| \le \alpha^t |e_0|$ for some $\alpha \in (0, 1)$. How many steps t are needed to make $|e_t| \le \varepsilon |e_0|$?

(b) Plug in $\alpha = |1 - 2\eta\sigma^2|$ and express t in terms of η, σ, and ε.

5 Two-Dimensional Diagonal System and Condition Number

Now consider a 2D system

$$\begin{bmatrix} \sigma_L & 0 \\ 0 & \sigma_S \end{bmatrix} \begin{bmatrix} w^{[1]} \\ w^{[2]} \end{bmatrix} = \begin{bmatrix} y^{[1]} \\ y^{[2]} \end{bmatrix},$$

with $\sigma_L \gg \sigma_S > 0$. We run gradient descent with a *single* learning rate η on the squared loss.

5.1 Mode-wise behavior

(a) Argue that each coordinate $w^{[1]}, w^{[2]}$ behaves like an independent scalar gradient descent on a quadratic with "σ" equal to σ_L and σ_S, respectively.

(b) Write the error recurrences $e_t^{(1)}$ and $e_t^{(2)}$ and the corresponding ratios $r_L(\eta)$ and $r_S(\eta)$.

5.2 Stability and limiting factor

(a) What η range must hold to keep *both* dimensions stable?

(b) Which σ (σ_L or σ_S) limits how large η can be while still converging?

5.3 Relative convergence speeds

(a) For a fixed η in the stable range, which component (associated with σ_L or σ_S) converges faster?

(b) Define a condition number $\kappa = \sigma_L^2/\sigma_S^2$. In one sentence, describe how κ affects the overall convergence rate.

6 Gradient Descent and SGD on Simple Models

You have N data points (x_i, y_i).

6.1 Constant model: $\hat{y}_i = \theta$

We want to minimize

$$L(\theta) = \frac{1}{N} \sum_{i=1}^{N} (y_i - \theta)^2.$$

(a) Compute $\dfrac{\partial L}{\partial \theta}$.

(b) Solve for the optimal $\theta^\star$ in closed form.

(c) Write the full-batch gradient descent update $\theta \leftarrow \theta - \alpha \dfrac{\partial L}{\partial \theta}$.

(d) Write the SGD update when you sample a single i uniformly from $\{1, \dots, N\}$.

(e) In one sentence, explain why each point has the same "vote" in the gradient.

6.2 Linear model: $\hat{y}_i = \theta x_i$

Now

$$L(\theta) = \frac{1}{N} \sum_{i=1}^{N} (y_i - \theta x_i)^2.$$

(a) Compute $\dfrac{\partial L}{\partial \theta}$.

(b) Solve for $\theta^\star$ in closed form.

(c) Write the full-batch gradient descent update for θ.

(d) Write the single-sample SGD update.

(e) Do all data points contribute equally in magnitude to the full gradient? Explain in terms of x_i.

7 Noise as Regularization

For each data point $\boldsymbol{x}_i \in \mathbb{R}^n$ we see noisy versions

$$\tilde{\boldsymbol{x}}_i = \boldsymbol{x}_i + \boldsymbol{N}_i, \quad \boldsymbol{N}_i \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 I_n) \text{ i.i.d.}$$

We consider minimizing the expected least-squares loss

$$\min_{\boldsymbol{w}} \mathbb{E}\big[\|\tilde{X}\boldsymbol{w} - \boldsymbol{y}\|_2^2\big].$$

7.1 One-dimensional sanity check

Let $x, y, w \in \mathbb{R}$, $\tilde{x} = x + N$, $N \sim \mathcal{N}(0, \sigma^2)$.

(a) Compute $\mathbb{E}\big[(\tilde{x}w - y)^2\big]$ explicitly by expanding and using $\mathbb{E}[N] = 0$ and $\mathbb{E}[N^2] = \sigma^2$.

(b) Show that this can be written as $(xw - y)^2 + (\text{something}) \cdot w^2$. Identify that "something" in terms of x and σ^2.

7.2 Matrix case

Let X be the $m \times n$ matrix of true inputs and $\tilde{X}$ the random noisy matrix with i-th row $\tilde{\boldsymbol{x}}_i^\top$.

(a) Write $\|\tilde{X}\boldsymbol{w} - \boldsymbol{y}\|_2^2$ as an inner product $(\tilde{X}\boldsymbol{w} - \boldsymbol{y})^\top(\tilde{X}\boldsymbol{w} - \boldsymbol{y})$.

(b) Take expectation and split into $\mathbb{E}[\tilde{X}^\top \tilde{X}]$ terms and cross terms, using that $\mathbb{E}[\boldsymbol{N}_i] = \boldsymbol{0}$ and $\operatorname{cov}(\boldsymbol{N}_i) = \sigma^2 I$.

(c) Show that $\mathbb{E}\big[\|\tilde{X}\boldsymbol{w} - \boldsymbol{y}\|_2^2\big]$ can be written as $\|X\boldsymbol{w} - \boldsymbol{y}\|_2^2 + \lambda\|\boldsymbol{w}\|_2^2$ for some λ. Identify λ in terms of σ^2 and possibly X.

7.3 Concept

(a) In words, why does adding isotropic Gaussian noise to inputs act like an ℓ_2 regularizer?

(b) If σ^2 increases, what happens to the effective λ?

8 General Tikhonov / Weighted Least Squares

Consider the optimization problem

$$\min_{\boldsymbol{x}} \ \|W_1(A\boldsymbol{x} - \boldsymbol{b})\|_2^2 + \|W_2(\boldsymbol{x} - \boldsymbol{c})\|_2^2,$$

where W_1, W_2, A are matrices and $\boldsymbol{x}, \boldsymbol{b}, \boldsymbol{c}$ are vectors.

8.1 1D toy version

Let $a, b, c, \alpha > 0$ be scalars and solve

$$\min_{x} \ (ax - b)^2 + \alpha(x - c)^2.$$

(a) Expand the expression and take the derivative with respect to x.

(b) Set the derivative to zero and solve for $x^\star$.

(c) Interpret $x^\star$ as a weighted average of b/a and c. What are the weights?

8.2 Matrix expansion

Back to the matrix problem:

(a) Write the objective as

$$(A\boldsymbol{x} - \boldsymbol{b})^\top W_1^\top W_1(A\boldsymbol{x} - \boldsymbol{b}) + (\boldsymbol{x} - \boldsymbol{c})^\top W_2^\top W_2(\boldsymbol{x} - \boldsymbol{c}).$$

(b) Take the gradient with respect to $\boldsymbol{x}$, set it to zero, and solve for $\boldsymbol{x}^\star$ in terms of A, $\boldsymbol{b}$, $\boldsymbol{c}$, W_1, and W_2.

8.3 Stacking trick

(a) Construct a matrix C and vector $\boldsymbol{d}$ such that

$$\|W_1(A\boldsymbol{x} - \boldsymbol{b})\|_2^2 + \|W_2(\boldsymbol{x} - \boldsymbol{c})\|_2^2 = \|C\boldsymbol{x} - \boldsymbol{d}\|_2^2.$$

Explicitly write C and $\boldsymbol{d}$ as a vertical concatenation.

(b) Using the OLS solution $\boldsymbol{x}^\star = (C^\top C)^{-1}C^\top \boldsymbol{d}$, verify that it matches the expression from 8.2.

(c) Choose W_1, W_2, and $\boldsymbol{c}$ so that this reduces exactly to standard ridge regression

$$\min_{\boldsymbol{x}} \; \|A\boldsymbol{x} - \boldsymbol{b}\|_2^2 + \lambda\|\boldsymbol{x}\|_2^2.$$

9 Gaussian MAP and Ridge Regression

Let $W \in \mathbb{R}^d$ and $Y \in \mathbb{R}^n$ be random vectors with

$$\begin{aligned} \text{Prior: } & W \sim \mathcal{N}(0, I_d), \\ \text{Noise: } & N \sim \mathcal{N}(0, I_n), \quad N \perp W, \\ \text{Observation model: } & Y = XW + \sqrt{\lambda}\,N. \end{aligned}$$

9.1 Covariances

(a) Compute $\Sigma_{WW} = \mathbb{E}[WW^\top]$.

(b) Compute $\Sigma_{YY} = \mathbb{E}[YY^\top]$ by expanding $Y = XW + \sqrt{\lambda}\,N$ and using independence.

(c) Compute $\Sigma_{WY} = \mathbb{E}[WY^\top]$.

9.2 Conditional expectation formula

For jointly zero-mean Gaussian (W, Y):

$$\mathbb{E}[W \mid Y = \boldsymbol{y}] = \Sigma_{WY}\Sigma_{YY}^{-1}\boldsymbol{y}.$$

(a) Plug your matrices from 9.1 into this formula and simplify $\mathbb{E}[W \mid Y = \boldsymbol{y}]$ into an expression involving X, λ, and $\boldsymbol{y}$ only.

(b) Show that this expression matches one of the standard ridge forms

$$\hat{\boldsymbol{w}} = (X^\top X + \lambda I)^{-1}X^\top \boldsymbol{y} \quad \text{or} \quad \hat{\boldsymbol{w}} = X^\top (XX^\top + \lambda I)^{-1}\boldsymbol{y}.$$

Indicate clearly which one you get and how to see it equals the other.

9.3 Interpretation

(a) Explain why the MAP estimate of W given $Y = \boldsymbol{y}$ is the same as the ridge regression solution.

(b) In words, how does λ control how much you "trust" the prior versus the data?